

AS/400 → Snowflake Migration Test Plan

1. Test Strategy Overview

The purpose of this test plan is to validate that all data migrated from **AS/400 (DB2/400)** to **Snowflake** is:

- **Accurate** (no corruption, truncation, or transformation errors)
- **Complete** (all rows, all columns, all tables)
- **Consistent** (matches legacy outputs)
- **Reconciled** (metrics, aggregates, and row-level checks)
- **Functional** (business rules from RPG/CL correctly rebuilt in DBT/Snowflake)
- **Performant** (ingestion, ELT, and queries meet SLAs)
- **Secure** (RBAC/ABAC enforced)

Testing spans **raw** → **normalized** → **staging** → **intermediate** → **marts**.

2. Test Phases & Scope

2.1 Data Extraction Testing

Validates correctness of data pulled from AS/400.

- PF/LF structure validation
- Encoding validation (EBCDIC → UTF-8)
- Packed/zoned decimal conversion
- Row counts & checksums
- CDC journaling validation

2.2 Landing Zone Testing

Ensures raw files landed correctly in S3/ADLS/GCS.

- File count validation

- File size comparison
- Metadata validation (timestamps, checksums)
- Folder structure validation

2.3 Snowflake Ingestion Testing

Validates COPY INTO, Snowpipe, Streams & Tasks.

- COPY INTO validation mode
- Rejected row analysis
- Snowpipe auto-ingest event validation
- CDC merge logic validation

2.4 Transformation Testing (DBT + ELT)

Ensures business logic is correctly rebuilt.

- DBT staging tests
- DBT intermediate logic tests
- DBT mart tests
- SCD Type 1/2 validation
- Referential integrity validation

2.5 Data Quality Testing

Covers completeness, accuracy, consistency.

- Null checks
- Domain checks
- Pattern checks
- Duplicate checks
- Outlier detection

2.6 Reconciliation Testing

The most critical phase for regulated industries.

- Schema validation
- Metric validation
- Row-level reconciliation
- Functional validation

2.7 Performance Testing

Ensures Snowflake meets SLAs.

- Ingestion throughput
- Query performance
- Warehouse sizing tests
- Clustering key effectiveness

2.8 Security & Access Testing

Validates Snowflake RBAC/ABAC.

- Role hierarchy validation
- Least-privilege access checks
- Masking policy validation
- Row-level security testing

3. Detailed Test Cases

3.1 Extraction Test Cases

- Validate PF/LF row counts match Snowflake raw zone
- Validate packed decimal → numeric conversion
- Validate EBCDIC → UTF-8 conversion

- Validate CDC journal entries match incremental loads

3.2 Landing Zone Test Cases

- Validate file names, counts, sizes
- Validate metadata files (checksums, timestamps)
- Validate folder structure

3.3 Ingestion Test Cases

- Validate COPY INTO row counts
- Validate rejected rows
- Validate Snowpipe event notifications
- Validate Streams & Tasks incremental merges

3.4 Transformation Test Cases

- Validate DBT staging column types
- Validate intermediate joins and logic
- Validate mart SCD behavior
- Validate surrogate key generation

3.5 Reconciliation Test Cases

- Schema: column count/type match
- Metrics: row counts, sums, distincts
- Row-level: hash comparison
- Functional: legacy vs Snowflake outputs

4. Entry & Exit Criteria

4.1 Entry Criteria

- AS/400 connectivity validated
- Landing zone provisioned
- Snowflake stages created
- DBT project initialized
- Test data available

4.2 Exit Criteria

- 100% schema validation passed
- 100% metric validation passed
- ≥ 99.99% row-level match (mortgage/financial standard)
- All critical defects resolved
- Business sign-off obtained

5. Roles & Responsibilities

Activity	Owner	
Test planning	PM / QA Lead	
Extraction testing	AS/400 SME + DE	
Ingestion testing	DE + Cloud Ops	
Transformation testing	DBT Lead + DA	
Reconciliation	QA + Business	
Security testing	Security Team	
Sign-off	Business + EA	

6. Defect Management

- Severity 1: Data corruption, missing data, incorrect logic
- Severity 2: Incorrect metrics, partial mismatches
- Severity 3: Non-critical formatting or metadata issues
- Severity 4: Cosmetic

Tracking via Jira, Azure DevOps, or ServiceNow.

7. Test Reporting

Daily & weekly reporting includes:

- Test execution progress
- Defect summary
- Reconciliation results
- SLA performance
- Go-live readiness score

8. Final Deliverables

- Test Plan (this document)
- Test Cases
- Test Scripts
- Reconciliation Reports
- Defect Logs
- Final Test Summary Report
- Business Sign-Off